AI/NLP Risk Engine with Portfolio Stress Testing

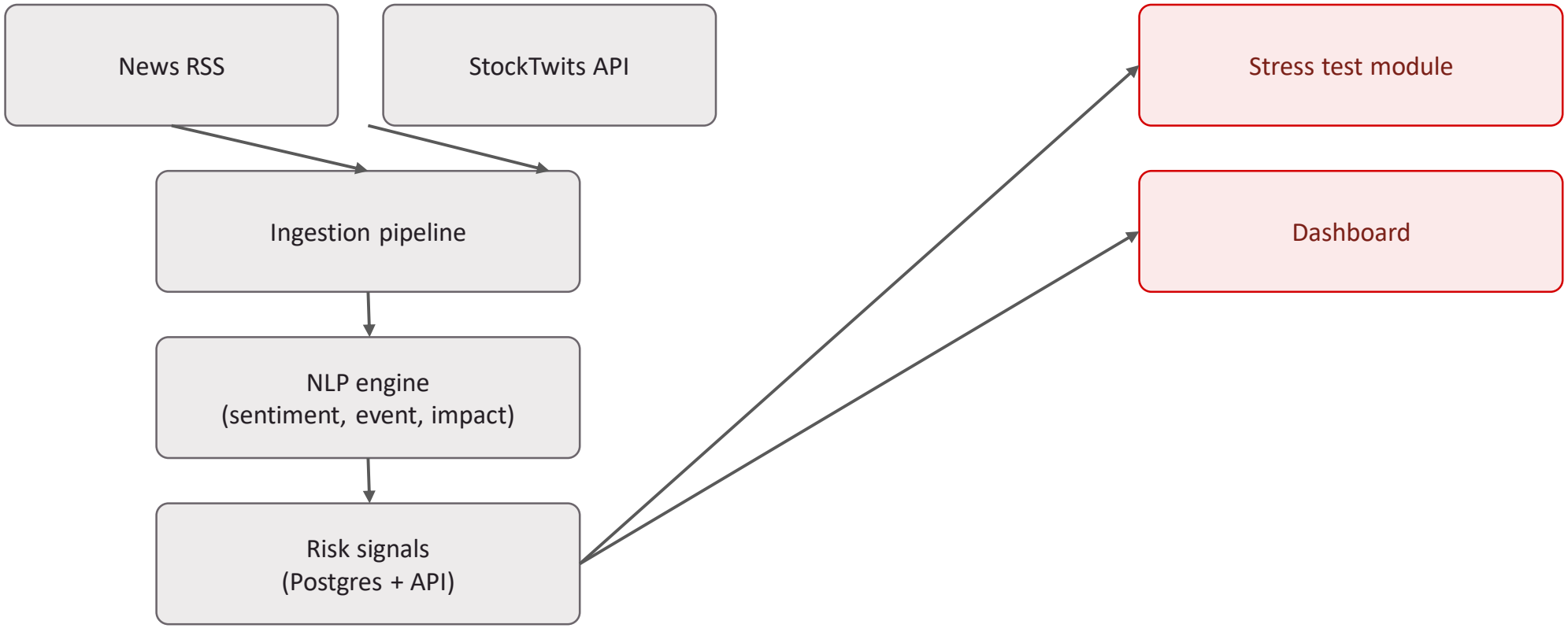
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Manual review of market-moving text cannot keep pace with how fast it arrives

- News and social chatter carry sentiment, event type, and severity, but a human analyst reading each item does not scale.
- Risk teams need these three fields as structured, queryable data, not prose.
- The open question: can an impact score be grounded in something checkable, instead of an unexplained LLM guess?

Text flows through three NLP stages before it ever reaches the portfolio model



Data sources: CNBC RSS (news); StockTwits public streaming API (social sentiment)

Grounding the impact score in retrieved precedent beats an ungrounded guess

	Ungrounded LLM guess	RAG-grounded (this system)
Basis for the score	Model's prior, unstated	3 nearest historical precedents
Traceable to evidence	No	Yes, precedents are shown
Defensible under Q&A	Weak	Each score cites its basis

- Reference set: 30 curated historical events, each tagged with an estimated severity.
- New text is embedded (ChromaDB + all-MiniLM-L6-v2) and matched to its 3 nearest neighbors.
- The impact score is a weighted average of those neighbors' known severities.

The engine stress-tests a portfolio only when the event is a genuine downside risk

Asset class	Shock applied	Effect
Equities	-10%	Loss
Bonds	-3%	Loss
Derivative (swap)	-5%	Loss
Loans	0%	Unchanged

Geopolitical event, impact score 8.5

-4.19%

portfolio value change under this shock

- Merger/Acquisition and Product Launch are excluded from this trigger, even at high impact scores.
- A stress-testing positive news was an early bug in this build, caught and fixed before submission.

Company tagging and event labels have known gaps worth stating plainly

Known limitations

- Company extraction is keyword matching against a fixed list, not full named-entity recognition.
- Zero-shot classification keeps some noise on short, terse headlines.
- The 30-event reference set is hand-curated, not a large historical corpus.

Planned fixes

- Replace keyword matching with NER-based company tagging.
- Backtest impact scores against real historical market reactions.
- Add a Monte Carlo layer for a loss distribution, not one point estimate.

Conclusions

- A working NLP engine turns news and social text into sentiment, event type, and a traceable impact score.
- A stress-test module reacts only to genuine downside events, verified end to end on a second machine.
- The limitations are known and documented, not hidden.

Repo: github.com/prateeknitcse/National-Institute-of-Technology-Arunachal-Pradesh-Prateek-Singh-Hackathon

Data sources: CNBC RSS, StockTwits public API. No proprietary or client data used.